ESE 3107 Concept Primer

Solar Thermal Engineering · KUET · read this first: the subject from zero, before the past-paper solutions

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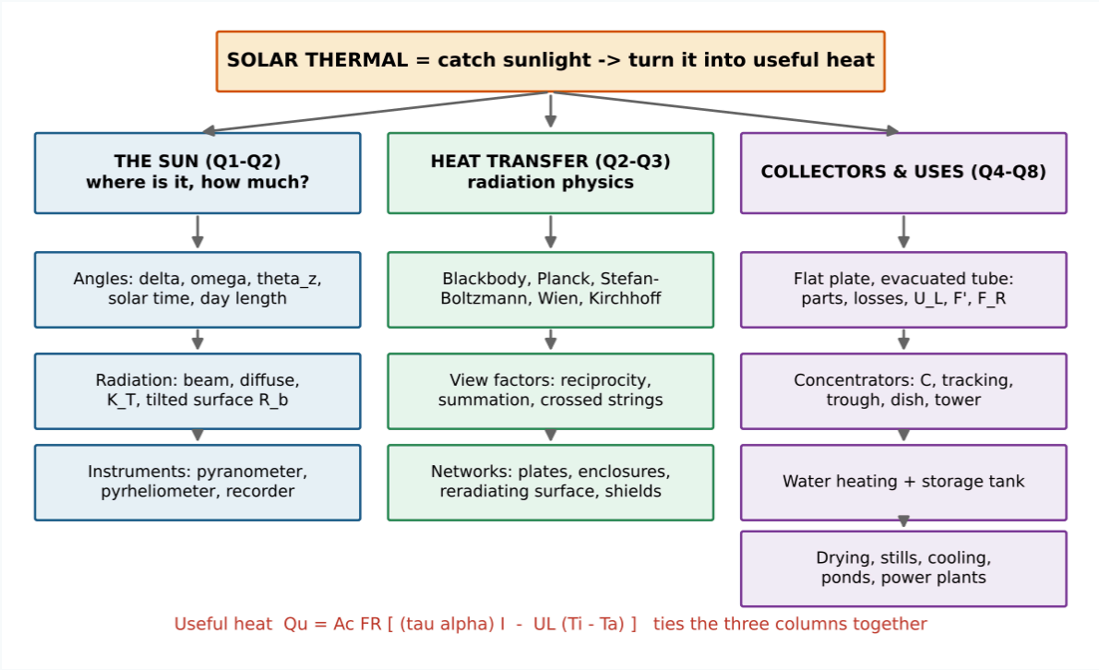
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How to use this primer

This booklet teaches solar thermal engineering from zero, in the order the exam needs it. Read it once before the Complete Solutions. Each section has a green **core idea** box (learn this first), the key terms, the formulas with every symbol and unit, and a short worked example. Section 2 is a maths refresher: most marks lost in this course come from degree/radian mix-ups, wrong signs of angles, and kelvin vs Celsius in radiation formulas.

The pass set (from the Question Map): Section A topics A1 (sun angles), A2 (solar radiation and instruments), A3 (radiation laws) and A5 (collectors); Section B topics B2 (concentrators), B3 (water heating and storage) and B4 (drying, stills, cooling, power). These cover Q1, Q2, Q4, Q6, Q7 and Q8, enough for three full questions in each section. A4 (view factors) and B1 (collector losses and U_L) are the next to add.

The whole course on one page



Concept map of ESE 3107

Read the map left to right: first find **where the sun is** and **how much radiation** reaches a surface; then learn the **physics of radiation** (how surfaces absorb, emit and exchange it); finally see how **collectors** turn it into heat and how that heat is **used** (hot water, drying, distillation, cooling, electricity).

Maths and units refresher

> Set your calculator to **degrees** for all sun-angle work. Use **kelvin** in every T^4 and in Planck's law. Keep 4 significant figures in intermediate steps.

Trigonometry you need

- sin, cos, tan of negative angles: $\sin(-x) = -\sin x$, $\cos(-x) = \cos x$, $\tan(-x) = -\tan x$. Southern declination and morning hour angles are negative.
- Inverse: if $\cos\omega_s = 0.159$ then $\omega_s = \cos^{-1}(0.159) = 80.85°$.
- Radians: π rad = 180°; in formulas with ω_s $(\pi/180)$ the angle is converted to radians.
- Small angles: $\sin\theta \approx \theta$ (rad) when $\theta < 1°$; $\sin(0.265°) = 0.004625$.

Day number n and time

Month	n for day i	Month	n for day i
Jan	i	Jul	181 + i
Feb	31 + i	Aug	212 + i
Mar	59 + i	Sep	243 + i
Apr	90 + i	Oct	273 + i
May	120 + i	Nov	304 + i
Jun	151 + i	Dec	334 + i

1 hour = 15° of hour angle; 1° of longitude = 4 minutes of time. Irradiance (W/m^2) × time (s) = irradiation (J/m^2); 1 MJ/m^2 in one hour = 277.8 W/m^2.

Hyperbolic functions and exponentials

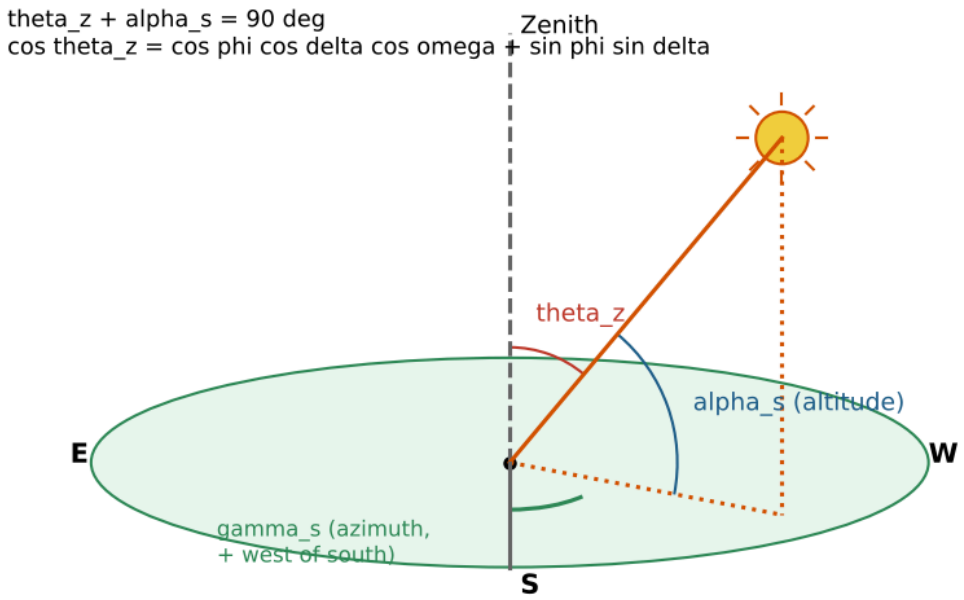
- $\tanh x = (e^x - e^{-x})/(e^x + e^{-x})$; for small x, $\tanh x \approx x$, so the fin efficiency $\tanh x/x \to 1$.
- $1 - e^{-x} \approx x - x^2/2$ for small x; $F'' = (1/x)(1 - e^{-x}) \approx 1 - x/2$.
- Fourth powers: $600^4 = 1.296 \times 10^{11}$, $\sigma 600^4 = 7348$ W/m^2; $\sigma 1000^4 = 56{,}700$ W/m^2.

Iteration

Some problems (top loss coefficient) need an unknown temperature to find the coefficients that give that temperature. Guess, compute, update, repeat until two answers agree (usually 2-3 rounds).

The sun and its angles (A1)

> The sun's direction at any moment is fixed by three things: **where** you are (latitude φ), **which day** it is (declination δ) and **what time** it is (hour angle ω). From these, the zenith angle θ_z (how high the sun is) and, for a tilted surface, the angle of incidence θ follow by spherical trigonometry.



Sun angles

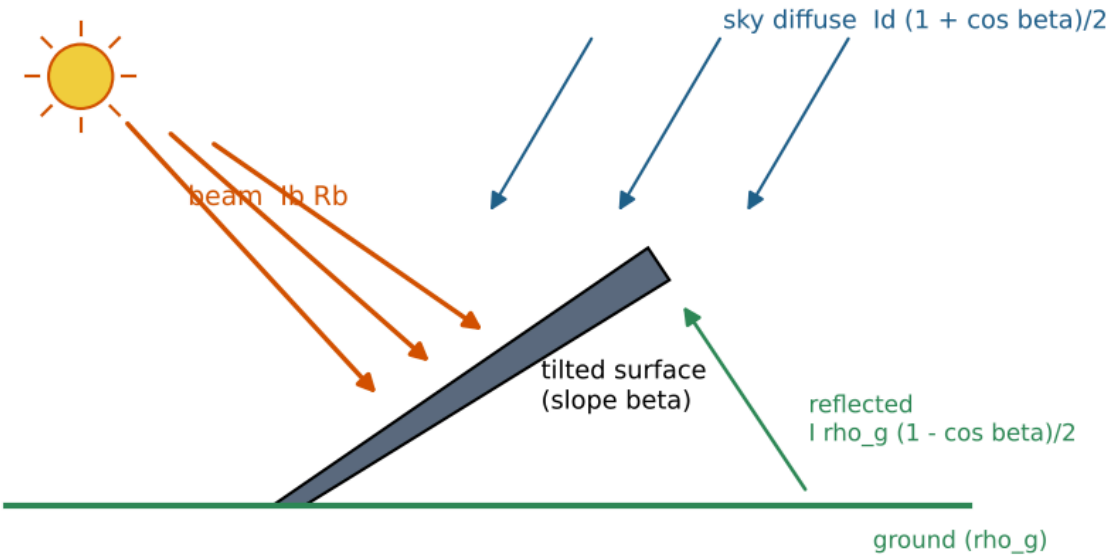
$\cos\theta_z = \cos\varphi\cos\delta\cos\omega + \sin\varphi\sin\delta;\quad \cos\omega_s = -\tan\varphi\tan\delta;\quad N = 2\omega_s/15$

Symbol	Meaning	Unit
φ	latitude, + north	deg
δ	declination	deg
ω	hour angle, 15°/h, morning −	deg
θ_z, α_s	zenith angle, altitude angle = 90 − θ_z	deg
γ_s, γ	solar and surface azimuth, 0 = south, west +	deg
β	surface slope	deg
n	day of the year	–
N	day length	h

Example: Khulna (φ = 22.8°) on January 18: δ = −20.73°, $\cos\omega_s$ = 0.159, ω_s = 80.85°, N = 10.78 h.

Solar radiation at the ground (A2)

Outside the atmosphere, 1367 W/m^2 arrives as a parallel beam. At the ground it is split into **beam** and **diffuse** parts; the clearer the sky (higher K_T), the more is beam. A tilted surface gets beam (times R_b), sky diffuse and ground-reflected radiation.



Radiation on a tilted surface

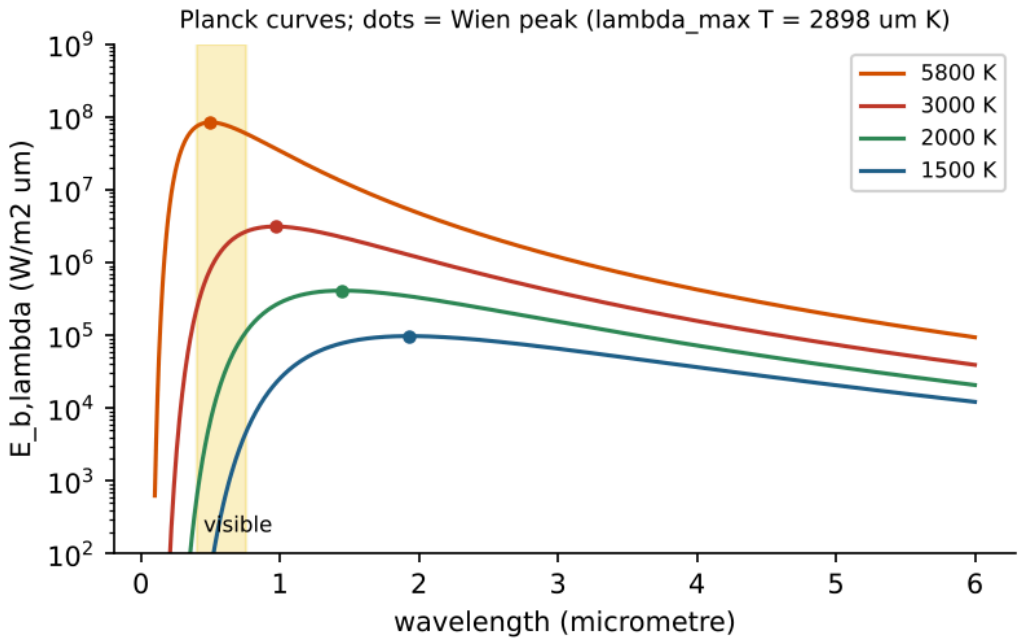
$$I_T = I_b R_b + I_d(1 + \cos\beta)/2 + I\rho_g(1 - \cos\beta)/2$$

Symbol	Meaning	Unit
G_{sc}	solar constant 1367	W/m^2
H, H_o	daily radiation at ground, extraterrestrial	MJ/m^2
K_T	clearness index H/H_o	–
I_b, I_d	beam, diffuse (horizontal)	W/m^2 or MJ/m^2h
R_b	geometric factor $\cos\theta/\cos\theta_z$	–
ρ_g	ground reflectance (albedo, ~0.2)	–

Example: I_b = 400, I_d = 150 W/m^2, R_b = 1.2, β = 25°, ρ_g = 0.2: I_T = 480 + 143.0 + 5.2 = 628 W/m^2.

Radiation physics (A3, A4)

Every surface emits $\varepsilon\sigma T^4$. A blackbody (ε = 1) is the ideal. Hot sources emit short waves, warm plates long waves (Wien). A good absorber is a good emitter at the same wavelength (Kirchhoff). Between surfaces, the exchange depends on the **view factor** and is solved with a resistance network.



Planck curves



surface resistance - space resistance - surface resistance; Q12 = (Eb1 - Eb2)/sum R

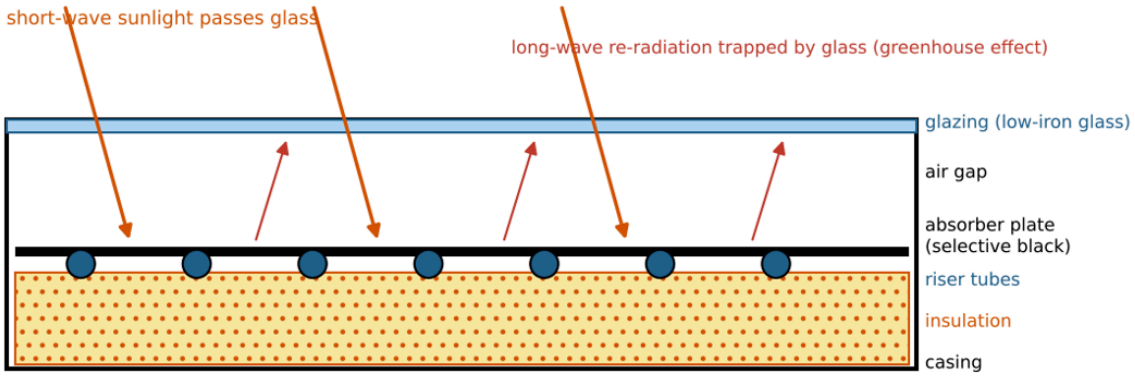
Two-surface network

Symbol	Meaning	Unit
σ	Stefan-Boltzmann constant 5.67×10^{-8}	W/m^2K^4
E_b	blackbody emissive power σT^4	W/m^2
I	intensity (per steradian)	W/m^2sr
$\varepsilon, \alpha, \rho, \tau$	emissivity, absorptivity, reflectivity, transmissivity	–
J, G	radiosity, irradiation	W/m^2
F_{ij}	view factor i to j	–
λ	wavelength	µm

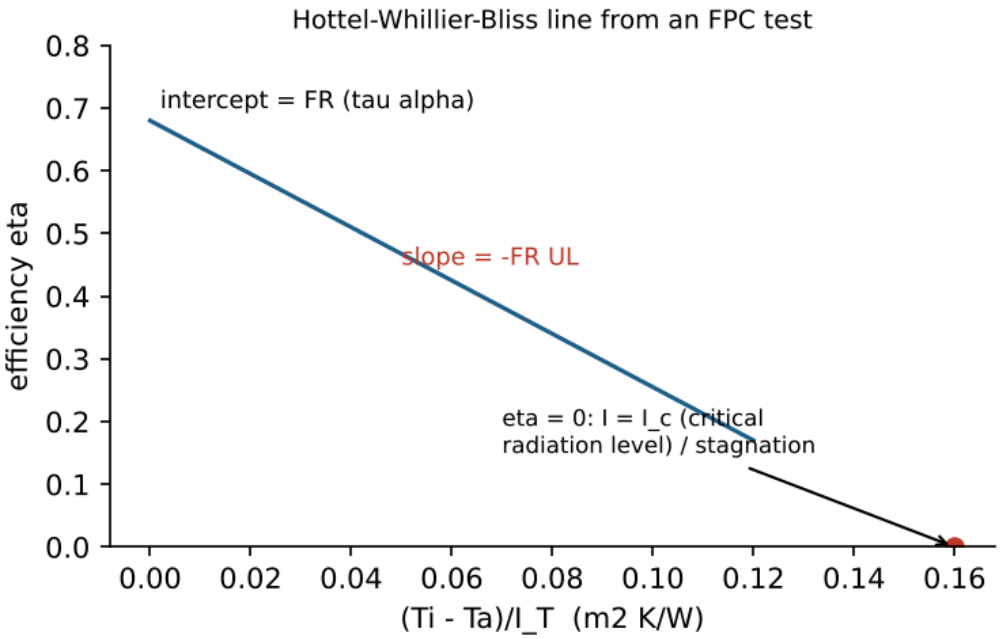
Example: plates at 900 K (ε 0.7) and 500 K (ε 0.5): $q = \sigma(900^4 - 500^4)/(1/0.7 + 1/0.5 - 1) = 13.9$ kW/m^2.

Solar collectors and their performance (A5, B1)

A flat plate collector is a black plate with tubes, under glass, in an insulated box. It absorbs $S = (\tau\alpha)I$ and loses $U_L(T_p - T_a)$; the difference is the useful heat. Written with the inlet temperature: $Q_u = A_c F_R[S - U_L(T_i - T_a)]$. An evacuated tube removes most of the loss.



Flat plate collector



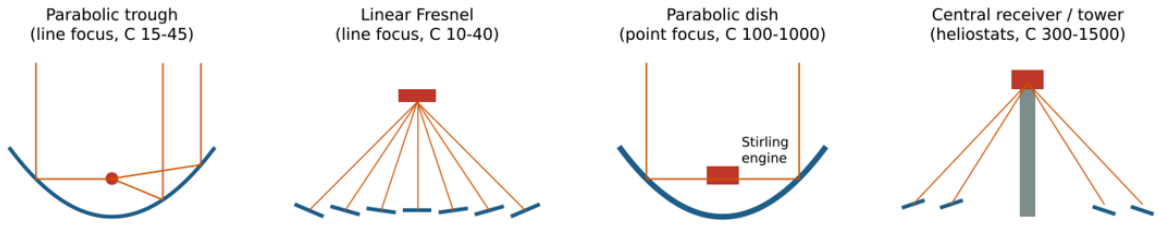
Efficiency line

Symbol	Meaning	Unit
A_c	collector area	m^2
S	absorbed solar $(\tau\alpha)I_T$	W/m^2
U_L, U_t, U_b, U_e	overall, top, bottom, edge loss coefficients	W/m^2K
F', F_R, F''	efficiency factor, heat removal factor, flow factor	–
F	fin efficiency	–
W, D, D_i	tube spacing, outer and inner diameter	m
h_{fi}	fluid-side heat transfer coefficient	W/m^2K
$\dot{m}$, c_p	mass flow, specific heat	kg/s, J/kg K
T_i, T_o, T_a, T_p	inlet, outlet, ambient, plate	°C

Example: 4 m^2, $F_R = 0.868$, $S = 640$ W/m^2, $U_L = 6.9$, $T_i - T_a = 5$: $Q_u = 4 \times 0.868 \times (640 - 34.5) = 2.10$ kW; $\eta = 2102/3200 = 66\%$.

Concentrating collectors (B2)

A mirror or lens of aperture A_a focuses beam radiation onto a small receiver A_r. The concentration ratio $C = A_a/A_r$ sets how hot the receiver can get; its upper limit $1/\sin^2\theta_s \approx 46{,}747$ comes from the sun's finite size and the second law. Concentrators must track the sun.



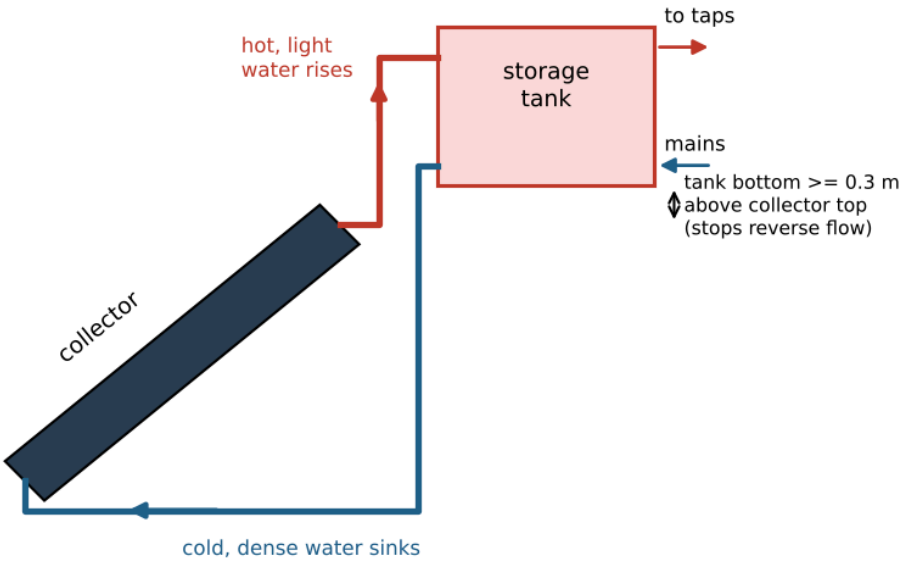
Concentrator types

Symbol	Meaning	Unit
C	concentration ratio A_a/A_r	–
$2\theta_c$	acceptance angle	deg
θ_s	half-angle of the sun ≈ 0.265°	deg
γ	intercept factor	–

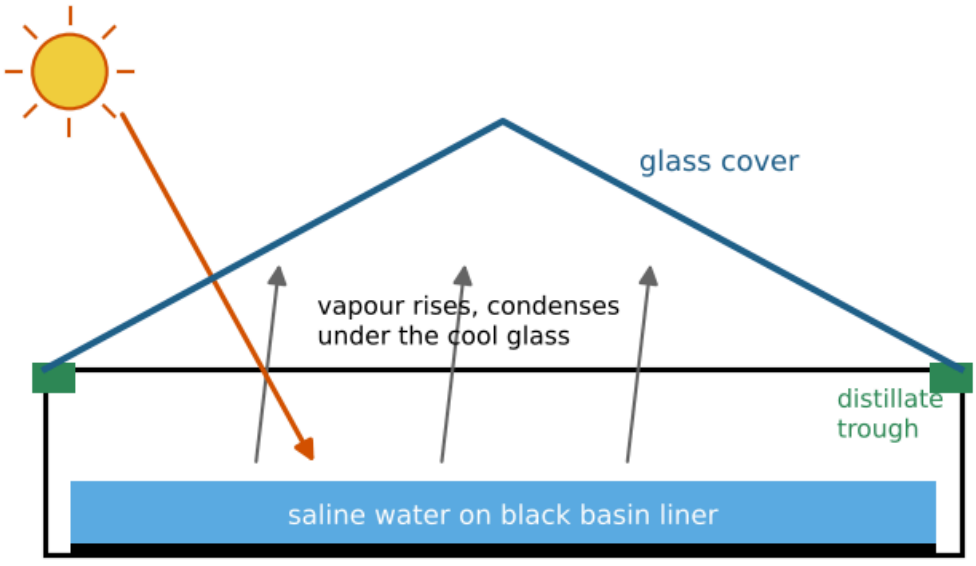
Example: a trough with 5 m aperture width and a 70 mm receiver tube (circumference 0.22 m): $C = 5/0.22 \approx 23$.

Water heating, storage and other uses (B3, B4)

Hot water rises: a tank above the collector gives free circulation (thermosiphon); otherwise a pump with a differential controller is used. A storage tank is followed hour by hour with an energy balance. Other uses: dryers (warm air), stills (evaporate and condense), desiccant/absorption cooling, and concentrators that drive turbines.



Thermosiphon water heater



Solar still

$T_s^+ = T_s + [Q_u - L - UA(T_s - T_a)\Delta t]/(mc_p)$; $q_{ew} = 0.016273 h_{cw}(P_w - P_g)$

Symbol	Meaning	Unit
Q_U, L	useful gain and load in one hour	MJ
UA	tank loss coefficient × area	W/K
m	mass of water in the tank	kg
h_{cw}	convective coefficient water-glass	W/m^2K
P_w, P_g	vapour pressure at water and glass temperature	Pa
λ	latent heat of evaporation	J/kg

Example: 500 kg tank at 45 °C, room 20 °C, UA = 12 W/K, no sun, load 12 MJ: loss = 12 × 3600 × 25 = 1.08 MJ; T = 45 − 13.08/2.09 = 38.7 °C after one hour.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
CPC	compound parabolic collector
CSP	concentrated solar power
DNI	direct normal irradiance
ETC	evacuated tube collector
FPC	flat plate collector
GHI	global horizontal irradiance
ICS	integrated collector storage
kWh	kilowatt-hour
LiBr	lithium bromide
LST	local standard time
MJ	megajoule
PV	photovoltaic
sr	steradian
ST	solar time
WMO	World Meteorological Organization